



Introduction to Material Processing and Advanced Materials Research Group

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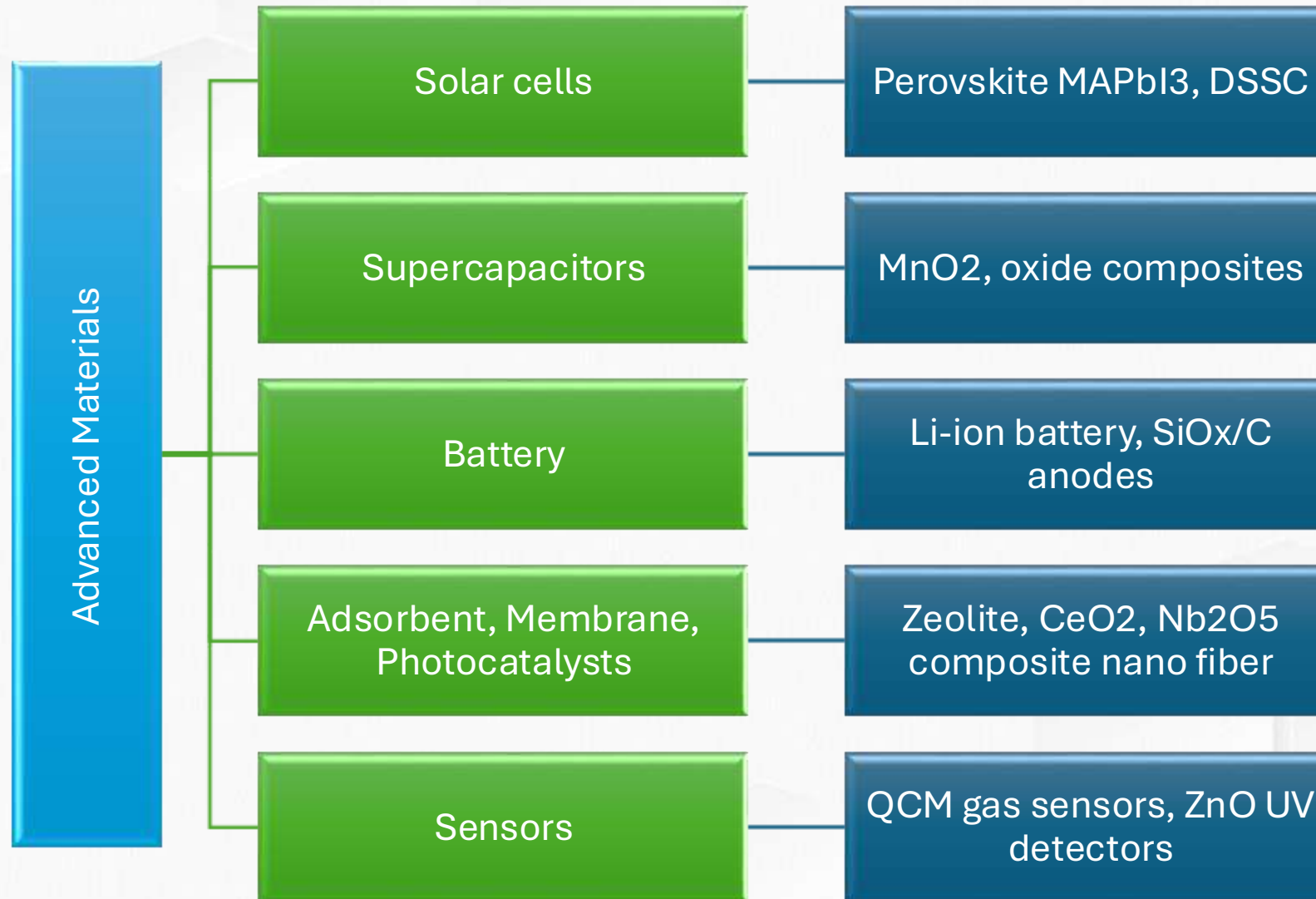
*Presented in AUT-ITERA Research Collaboration Webinar
South Lampung, 4th December 2024*

Introduction: Specialization Areas

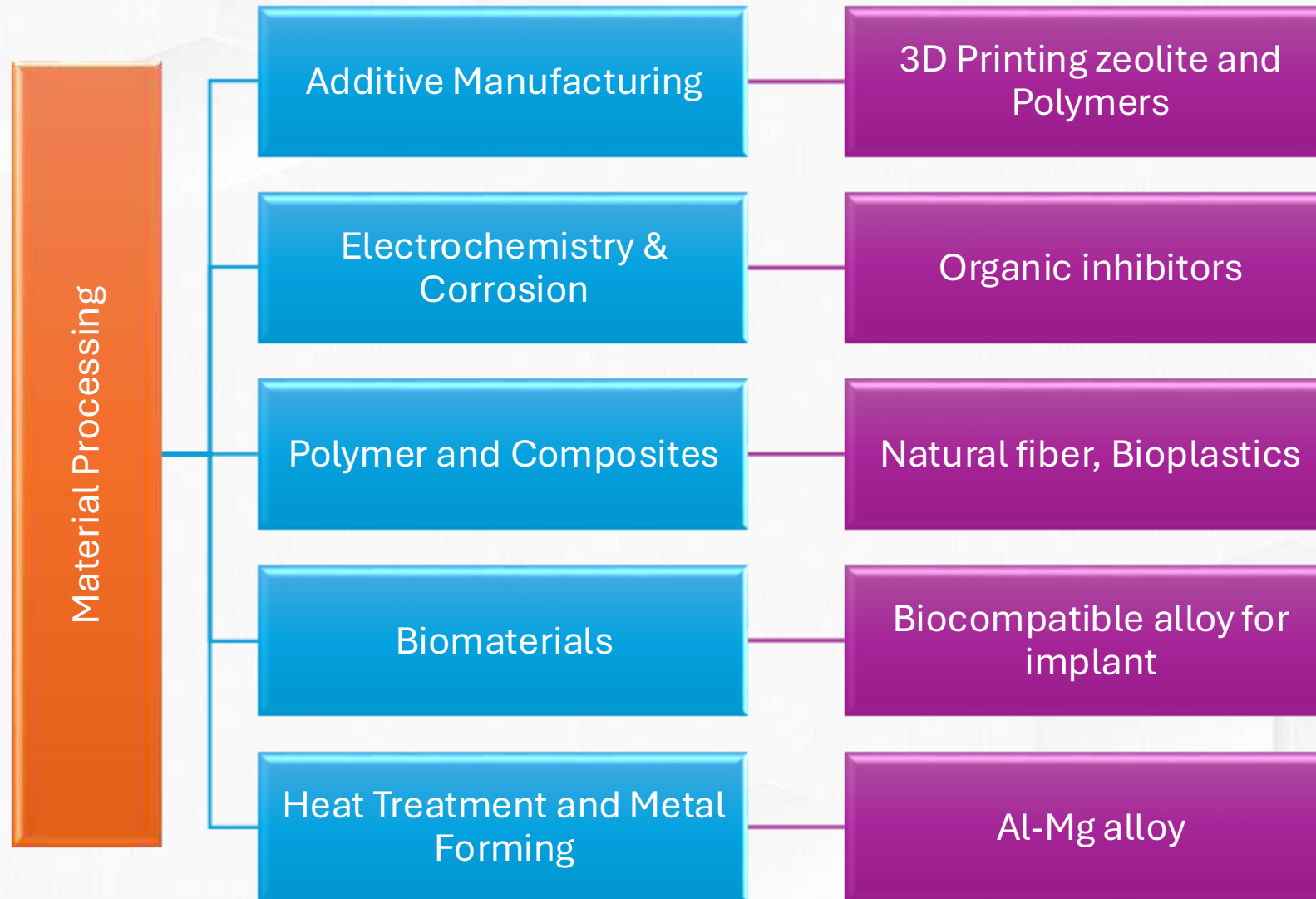
“Material Processing and Advanced Materials” Research Group

No.	Name	Speciality
1	Dr. Aditya Rianjanu , <u>S.Si.</u> (Head of RG)	Nanomaterials for Contaminant Detection and Environmental Remediation
2	Dr. Muhamad F. Arif , S.T., M.Sc.	Additive Manufacturing
3	Dr. Nur I. Khamidy , S.T., M.Sc.	Materials for Electrochemical Systems
4	Dr. Eka Nurfani , <u>S.Si.</u> , <u>M.Si.</u>	Materials for energy harvesting
5	Dr. Eng. Bambang A. Wahjoedi , M.Sc. Tech.	Electrochemistry & Corrosion
6	Dr. Ahmad Afandi	Alloy Design
7	Wahyu S. Sipahutar , <u>S.Si.</u> , <u>M.Sc.</u>	Composite Organic Materials for Optoelectronics
8	Agita Phasa , S.T., M.T.	Material Computation
9	Anisa Fitri , S.T., M.T.	Materials for Biomedicine
10	Bayu Prasetya , <u>S.Si.</u> , M.T.	Ceramic Materials for Electronic Devices
11	Istiara R. Hanifah , <u>S.Si.</u> , M.T.	Polymer Materials and Material Chemistry
12	Mhd. Yasin Siregar , S.T., M.T.	Extractive Metallurgy
13	Mochammad G. I. Khan , S.T., <u>M.Sc.</u>	Heat Treatment and Metal Forming
14	Quratul Aini , <u>S.Si.</u> , M.T.	Materials for Energy Storage

Research Focus - Advanced Materials

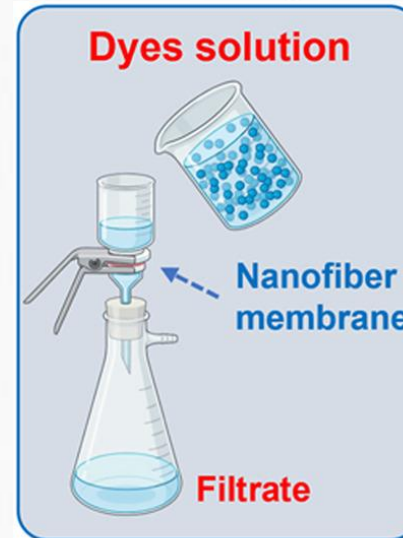
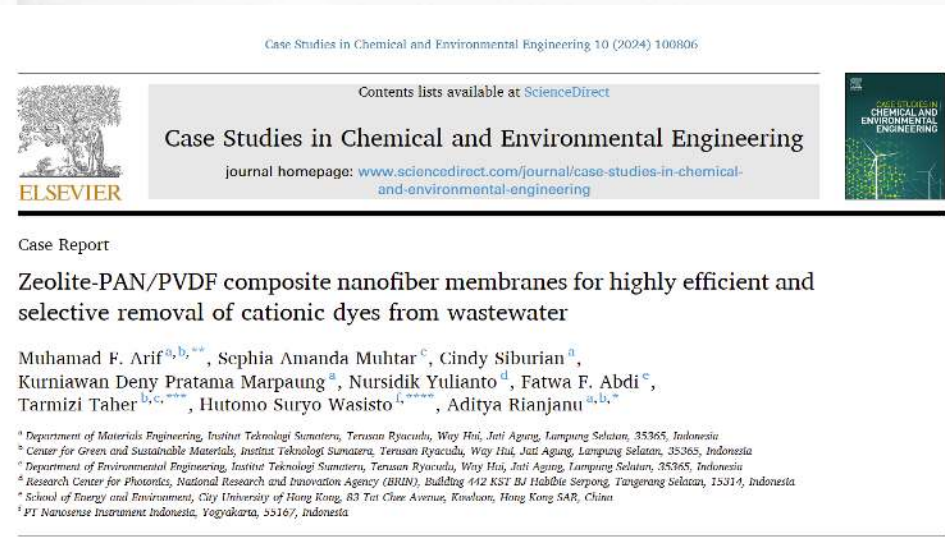


Research Focus - Material Processing

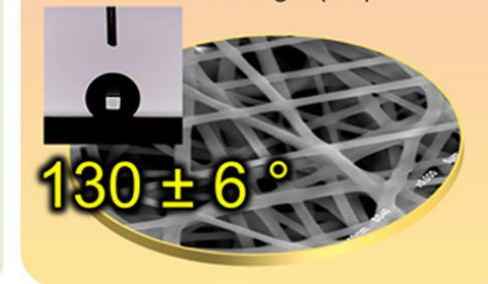
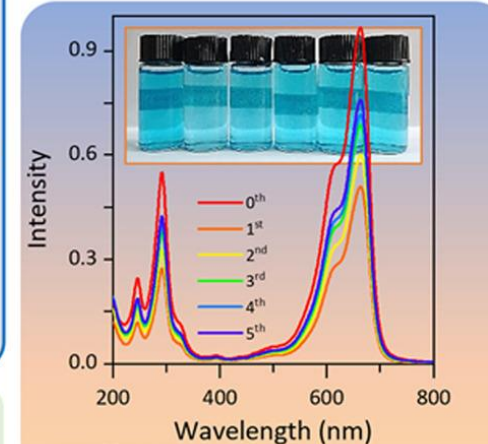


Highlight (2024) – Membranes

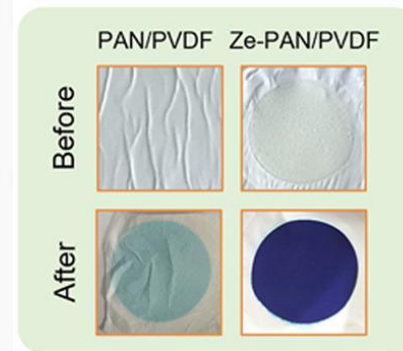
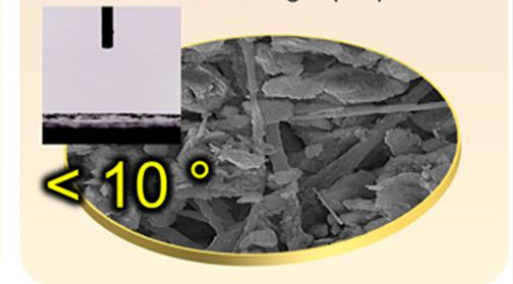
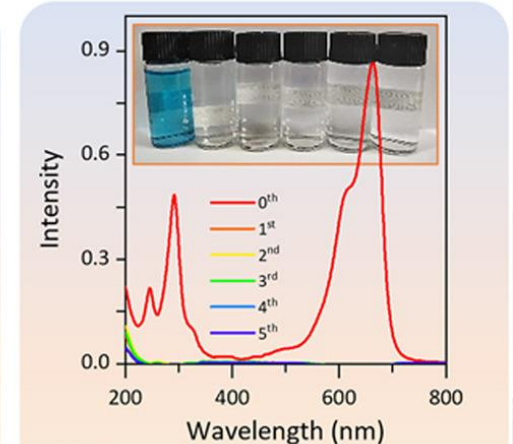
PAN (Polyacrylonitrile)/PVDF (Polyvinylidene Fluoride)



PAN/PVDF nanofiber

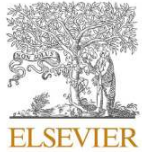


Zeolite-PAN/PVDF nanofiber



Highlight (2024) – Photocatalysts

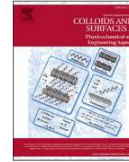
Colloids and Surfaces A: Physicochemical and Engineering Aspects 682 (2024) 132919



Contents lists available at ScienceDirect

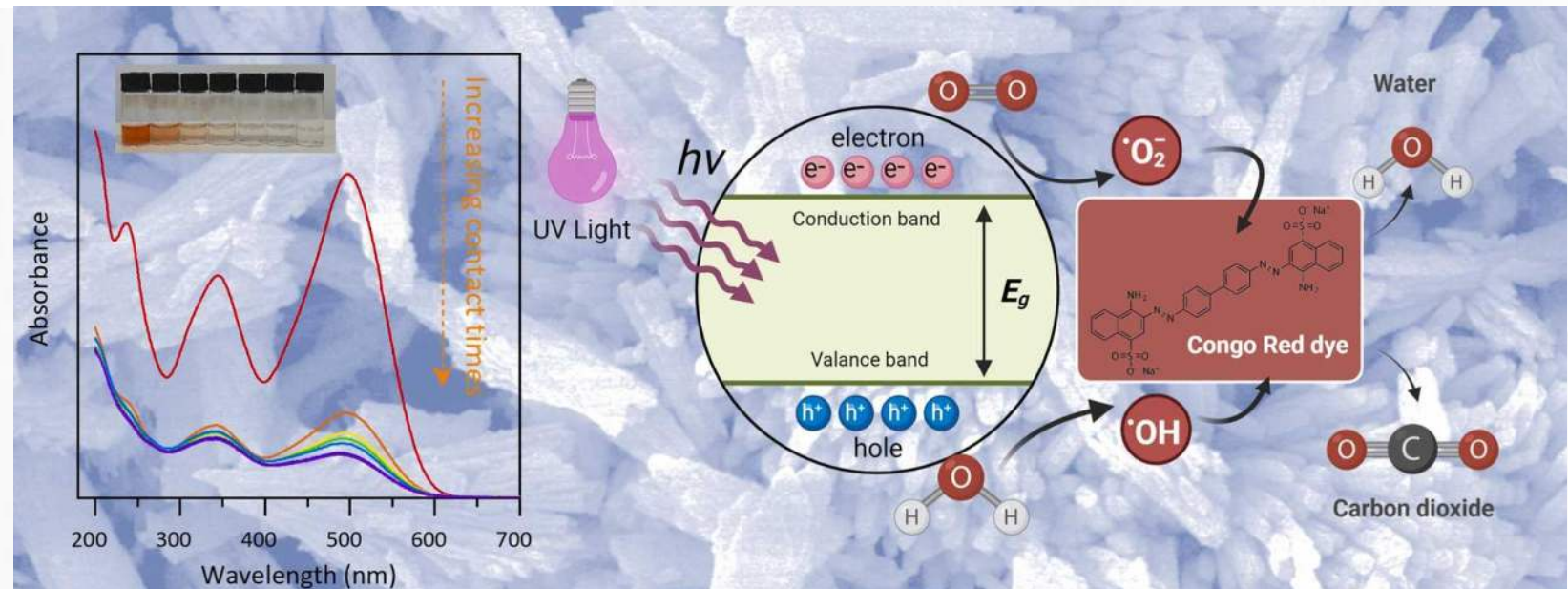
Colloids and Surfaces A: Physicochemical and
Engineering Aspects

journal homepage: www.elsevier.com/locate/colsurfa



Photocatalytic degradation of aqueous Congo red dye pollutants by rare-earth metal oxide (CeO₂) nanorods

Aditya Rianjanu^{a,b,*}, Aisyah Sabilla Putri Mustamin^a, Elisabeth Kartini Arum Melati^{b,c}, Rizky Aflaha^d, Nur Istiqomah Khamidy^a, Maisari Utami^e, Khairurrijal Khairurrijal^{b,f,i}, Kuwat Triyana^d, Fatwa F. Abdi^g, Hutomo Suryo Wasisto^h, Tarmizi Taher^{b,c,**}



Highlight (2024) – 3D Printing

Results in Surfaces and Interfaces 17 (2024) 100337

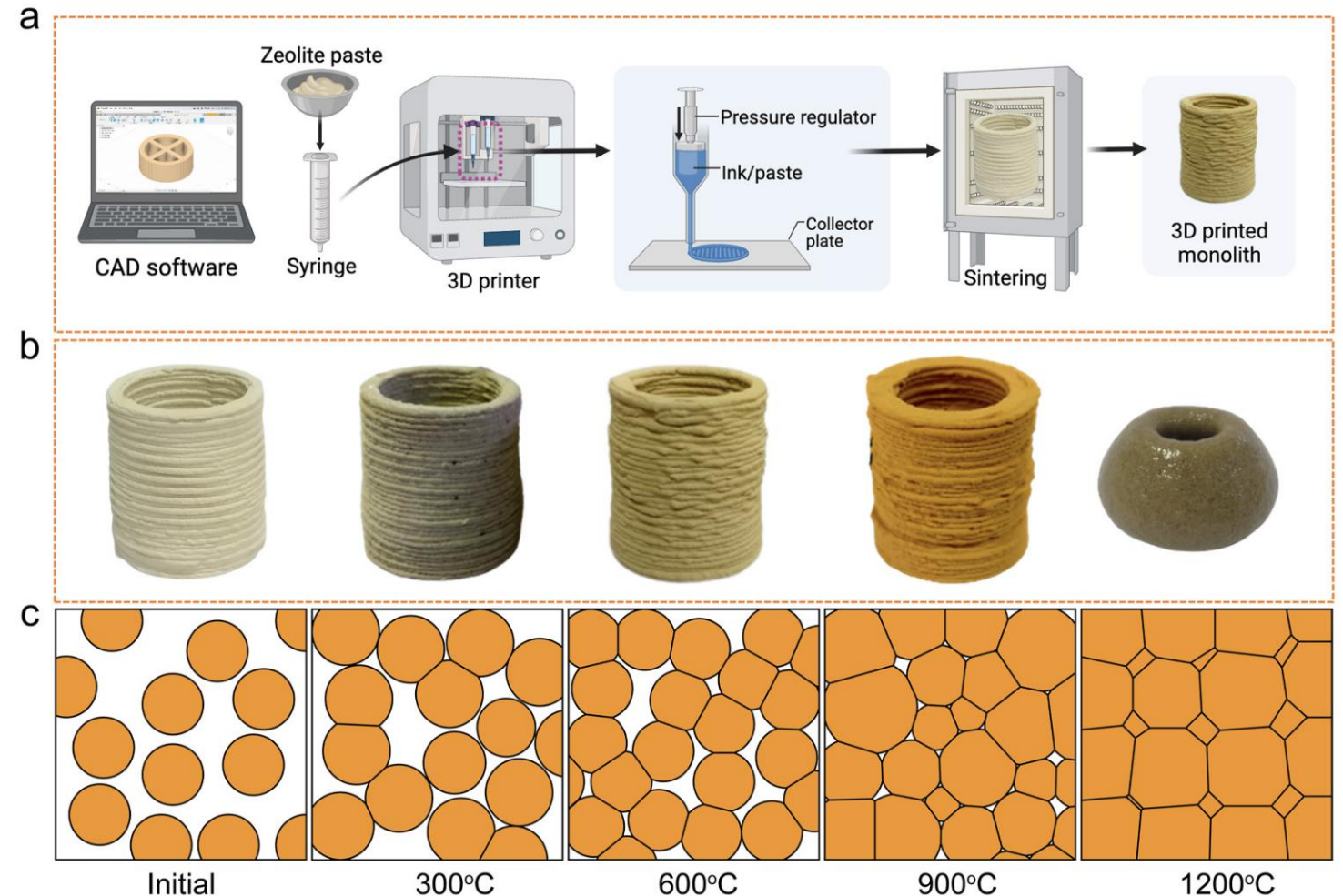


Contents lists available at ScienceDirect
Results in Surfaces and Interfaces
journal homepage: www.elsevier.com/locate/rsurfi



Examining the influence of sintering temperatures on the efficiency of 3D-printed natural zeolite for methylene blue dye adsorption

Aditya Rianjanu^{a,b}, Tarmizi Taher^{b,c}, Fuji Desriani^a, Rini O. Delmita^a, Audrey G.N. Sianturi^a,
Sephia Amanda Muhtar^c, Bambang Ariwahjoedi^a, Nur Istiqomah Khamidy^a,
Damar Rastri Adhika^{d,e}, Muhamad F. Arif^{a,b,*}



Cellulose Impact on Bioplastic Performance: A Study on Mechanical Strength, Physical Properties, and Degradation of Water Hyacinth and Kepok Banana Peel-derived Materials

Faiza Armalia Putri^(1,a), Aditya Rianjanu^(1,b) and Wahyu Solafide Sipahutar^{(1,c)*}

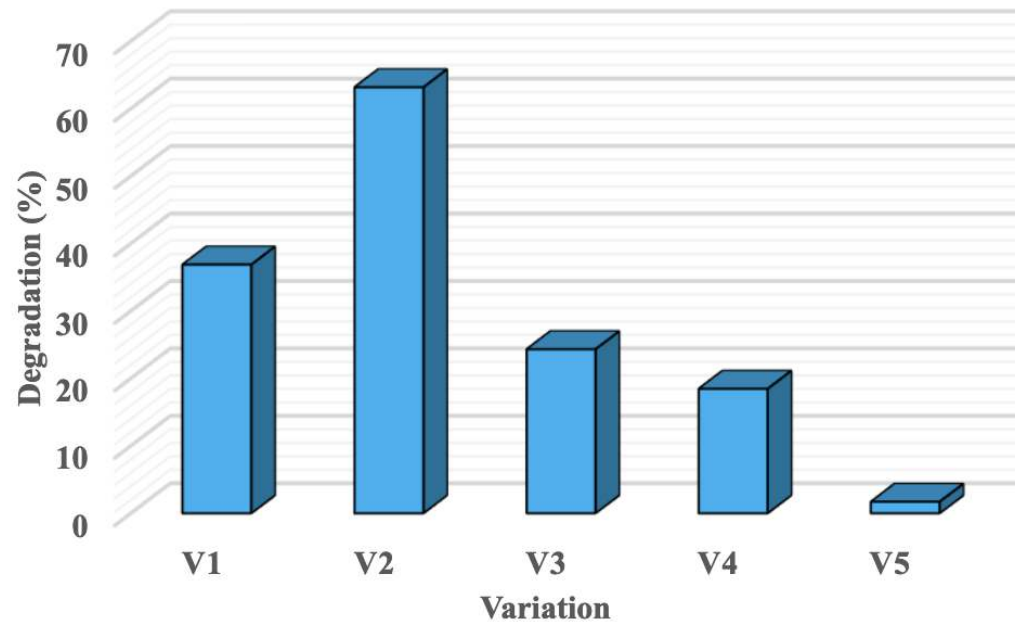


Figure 2. Bioplastics degradation rate (%)

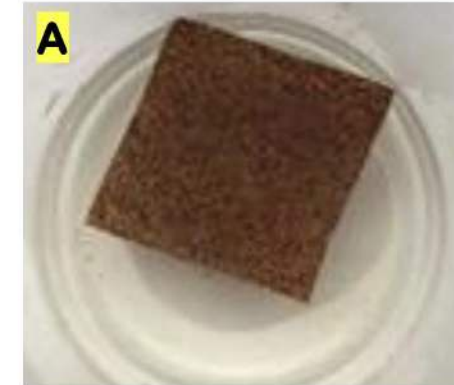


Figure 4.a) Bioplastics immersed in water
b) The result of bioplastics after water absorption testing

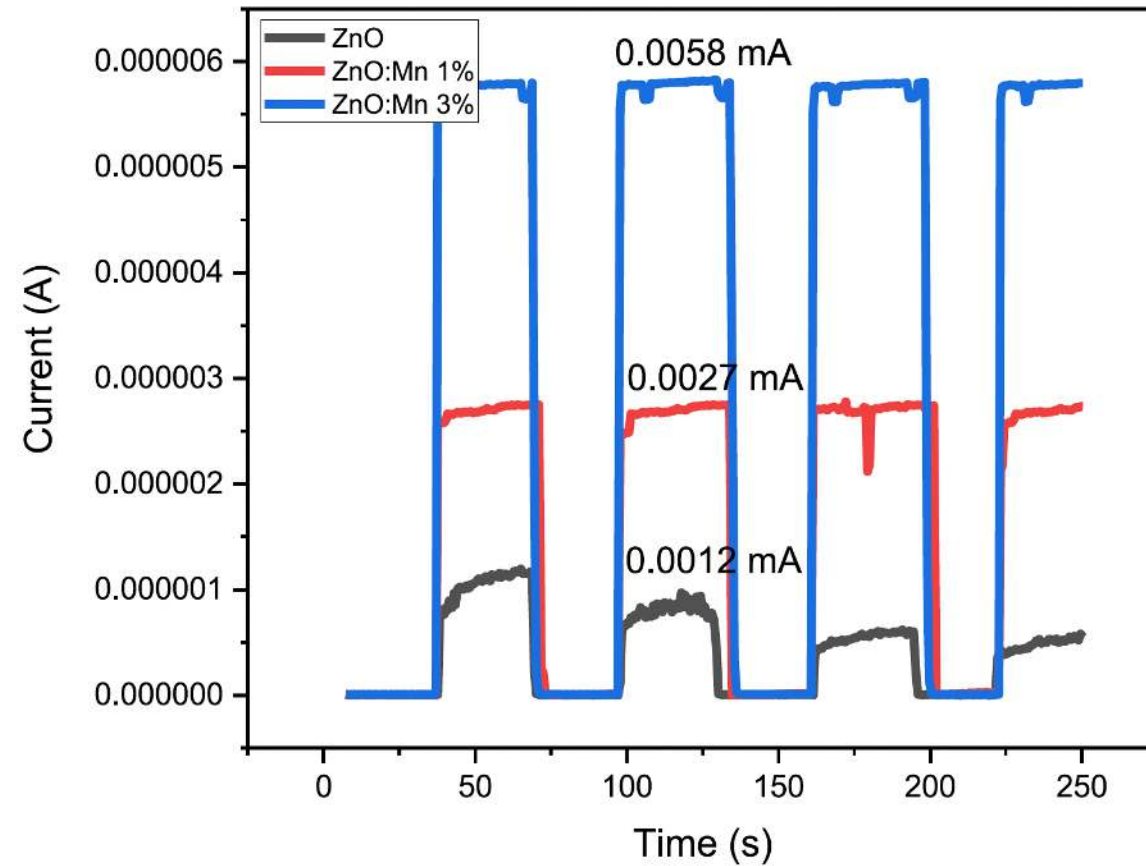


Fig. 5. Characterization of I-T curves at a bias voltage of 0V for Mn-doped ZnO films.

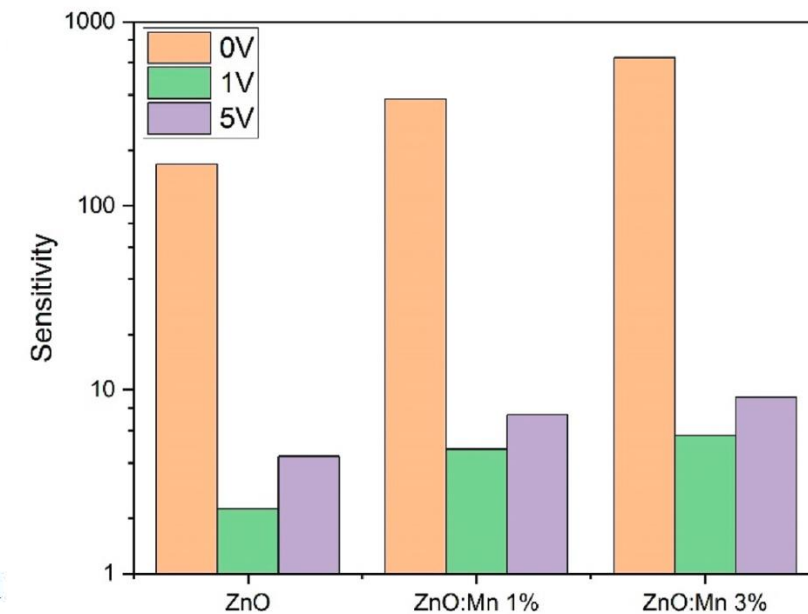


Fig. 8. Calculated sensitivity of Mn-doped ZnO films on p-Si substrate at the bias voltage of 0, 1, and 5 V.

Journal of Electronic Materials
https://doi.org/10.1007/s11664-024-11386-1

ORIGINAL RESEARCH ARTICLE

Enhancing the Performance of MAPbI₃-Based Perovskite Solar Cells Fabricated Under Ambient Air: Effect of Cu, Ni, and Zn Doping into TiO₂

Mezan Adly Al Qadri¹ · Wahyu Solafide Sipahutar¹ · Nur Istiqomah Khamidy¹ · Iwan Syahjoko Saputra² · Eri Widiyanto³ · Widi Astuti⁴ · Eka Nurfani¹

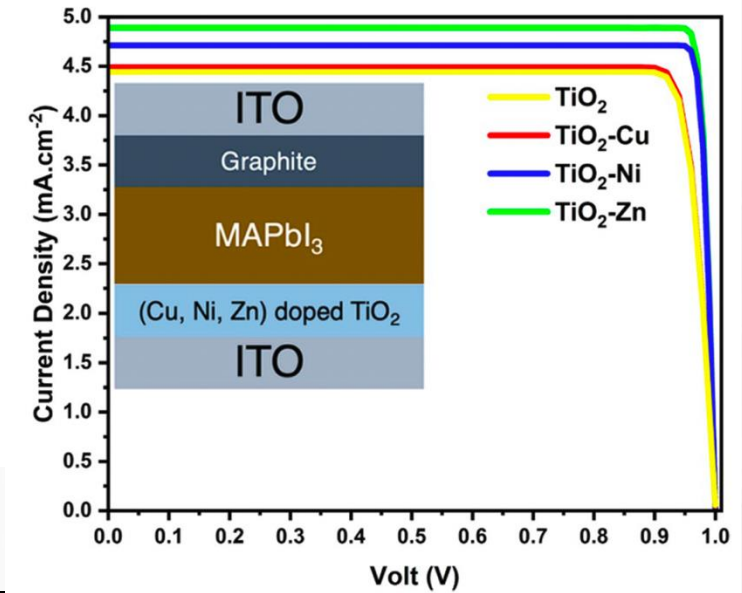


Fig. 6 J - V curves of fabricated MAPbI₃-based PSC devices.

Table IV Photovoltaic characteristics of MAPbI₃-based PSCs

Samples	V_{OC} (V)	V_{mp} (V)	I_{mp} (A)	I_{SC} (A)	J_{SC} (mA cm ⁻²)	Fill factor (%)	P_{max} (VA)	PCE (%)
TiO ₂	0.99	0.92	8.09×10^{-3}	0.01012	4.440	75	0.007	5.74
TiO ₂ :Ni	1.01	0.95	1.04×10^{-2}	0.01060	4.711	85	0.009	7.63
TiO ₂ :Cu	1.09	0.85	9.80×10^{-3}	0.01019	4.489	82	0.008	6.43
TiO ₂ :Zn	1.10	0.99	1.09×10^{-2}	0.01160	4.9	89	0.010	8.35

V_{OC} = Open-circuit voltage, V_{mp} = Voltage at maximum power, I_{mp} = Current at maximum power, I_{SC} = Short-circuit current, J_{SC} = Short circuit current density, FF = Fill factor, P_{max} = Maximum power point, PCE = Power conversion efficiency

Highlight (2024) - Battery

New Insights into Phytolith Characteristics in Wild Grass as the Source of Sustainable Silicon Oxide Anodes

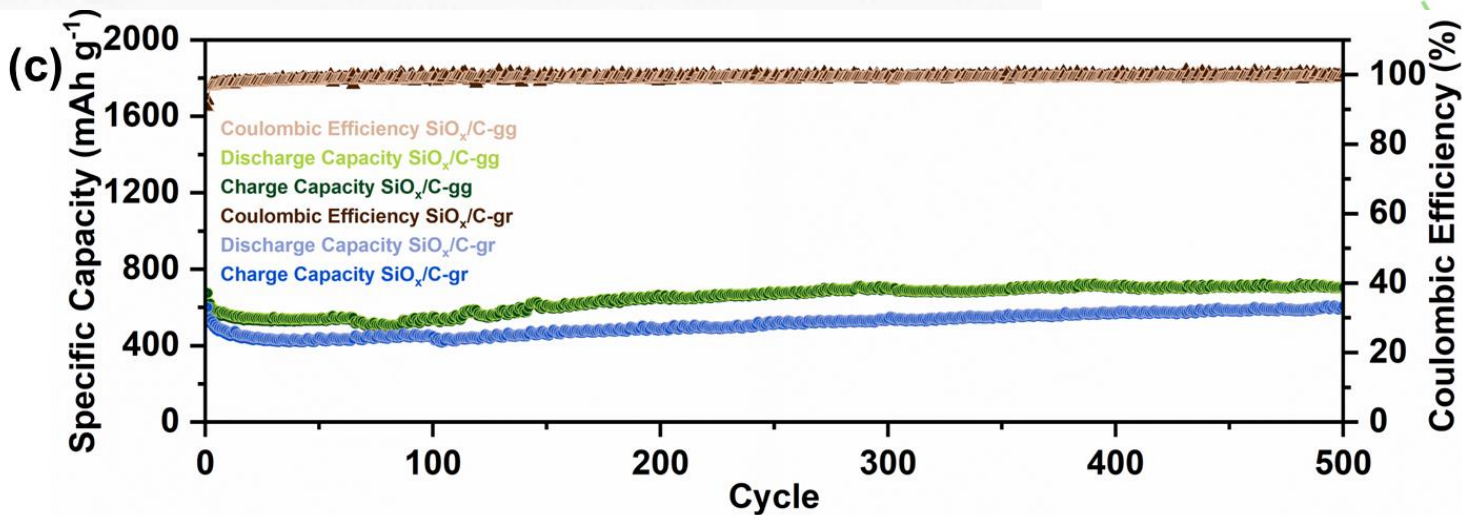
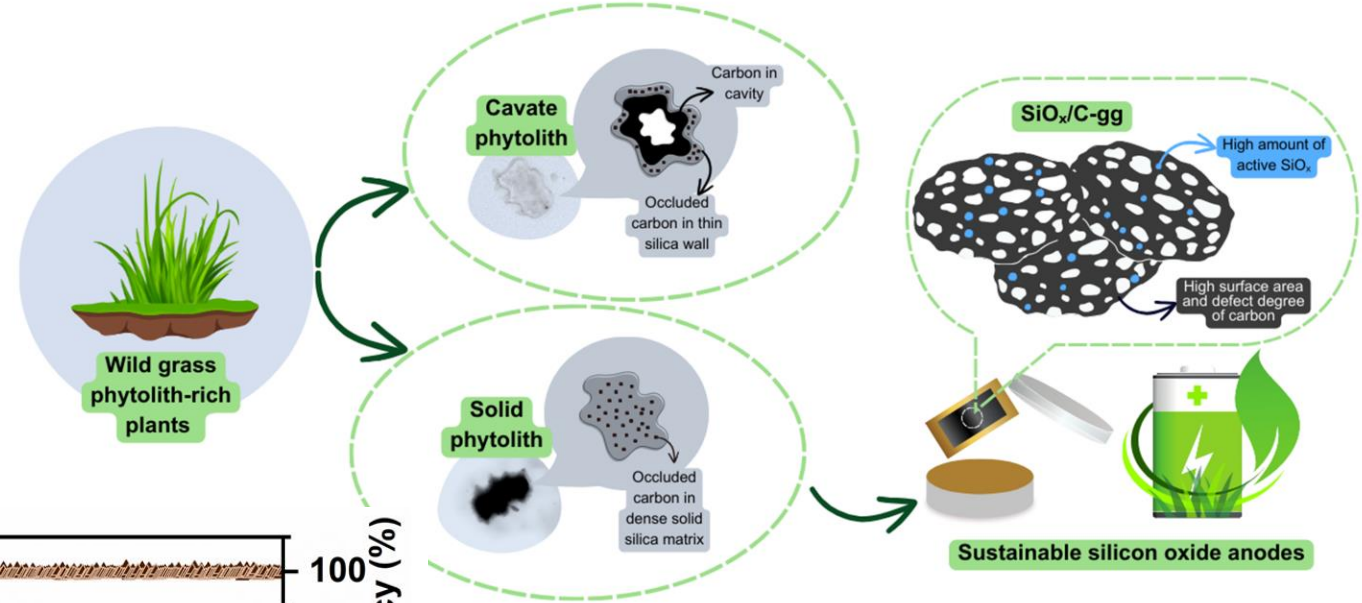
Quratul Aini, Pilar Bela Persada, M. Hamzah Raihan Pasha, Naufal Hanif Hawari, and Afriyanti Sumboja*



Cite This: <https://doi.org/10.1021/acssuschemeng.4c05138>

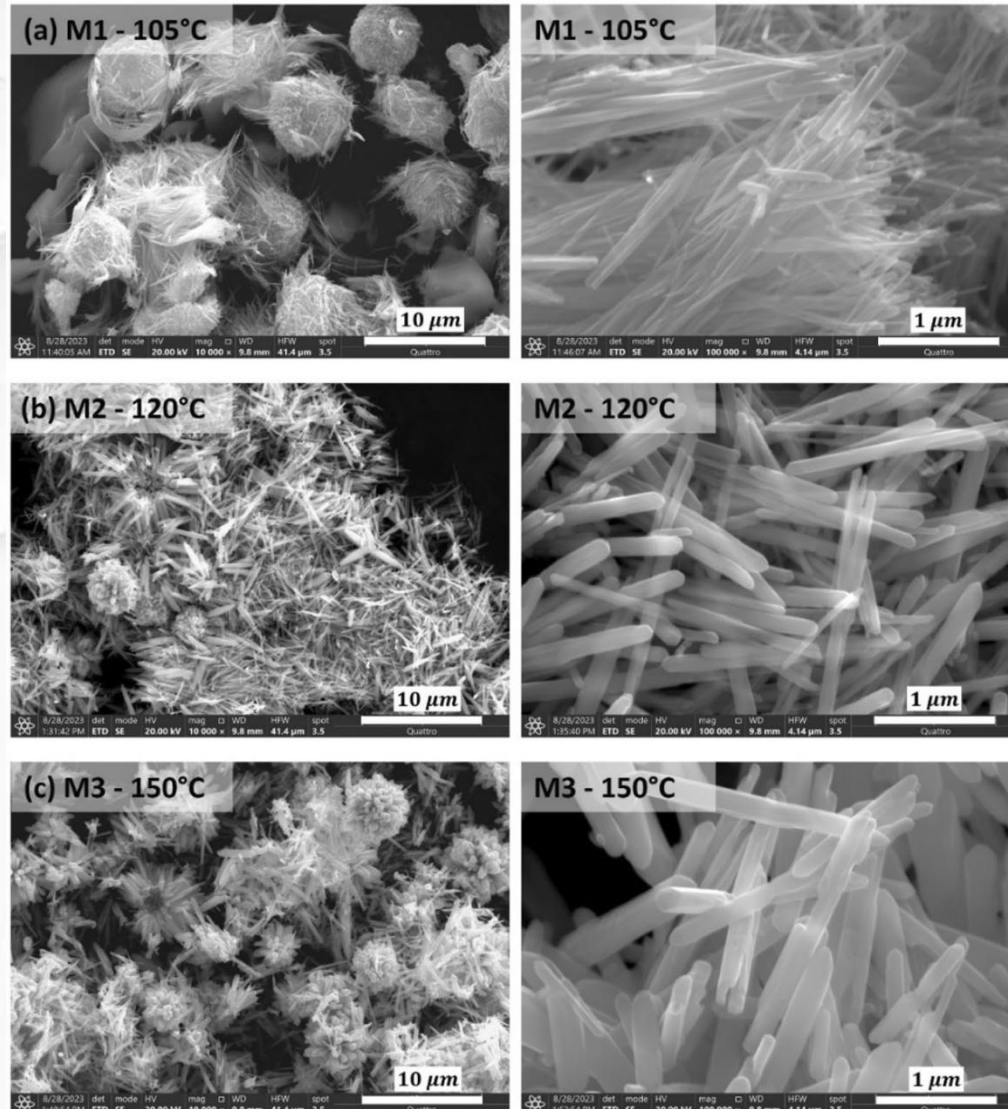


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100% CE with 704 mAh g^{-1}
discharge capacity after
500 cycles

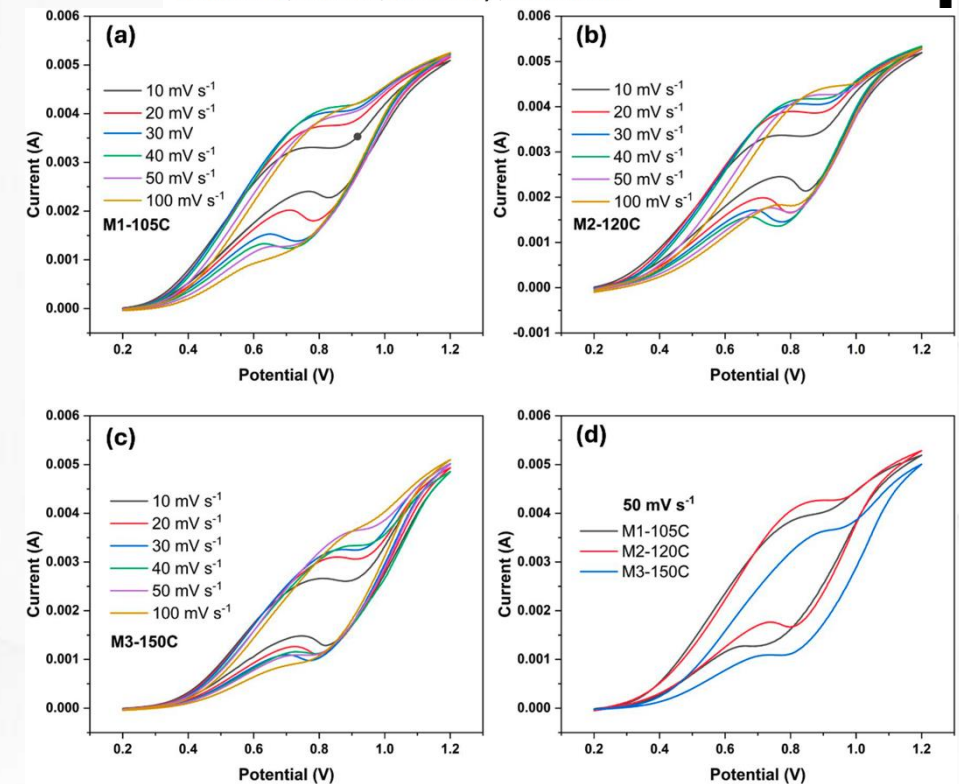
Highlight (2024) - Supercapacitors



J Mater Sci: Mater Electron (2024) 35:2046

Effect of hydrothermal temperature on the structural and electrochemical properties of MnO₂-based supercapacitors

Eka Nurfani^{1,*}, Paulus Fau¹, Nur I. Khamidy¹, and Resti Marlina²



Research collaboration



Thank you 😊